

$$A_\mu = (-\phi, \vec{A}) \quad \eta^{\mu\nu} = (\eta^{-1})_{\mu\nu} \quad \eta_\mu = \eta^{\mu\nu} = \begin{pmatrix} -1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{pmatrix}$$

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \quad A_\mu \rightarrow \partial_\mu \lambda + A_\mu \quad F'_{\mu\nu} = \partial_\mu (A_\nu + \partial_\nu \lambda) - \partial_\nu (\partial_\mu \lambda + A_\mu) = F_{\mu\nu}$$

$$A^0 = 0 \quad \nabla \cdot \vec{A} = \partial_i A_i = 0$$

$$\eta^{\mu\nu} \partial_\nu A_\mu = 0 \Rightarrow \nabla \cdot \vec{A} + \partial_t \phi = 0 \quad c=1$$

$$(\nabla \times \vec{B})^i = \frac{\partial E^i}{\partial t} + 4\pi J^i \quad \vec{B} = \nabla \times \vec{A} \quad E = -\partial_t \phi - \partial_i A_i$$

$$\nabla \times (\nabla \times \vec{A}) = \frac{\partial}{\partial t} (-\nabla \phi - \frac{\partial \vec{A}}{\partial t}) = 4\pi \vec{J} \quad \square = \partial_t^2 - \nabla^2$$

$$\nabla (\nabla \cdot \vec{A}) - \nabla^2 \vec{A} = \frac{\partial}{\partial t} (\nabla \phi - \frac{\partial \vec{A}}{\partial t}) = \nabla (\underbrace{\nabla \cdot \vec{A} + \partial_t \phi}_0) - \nabla^2 \vec{A} + \partial_t^2 \vec{A} = 4\pi \vec{J} \rightarrow \square A^i = 4\pi J^i$$

$$\nabla \cdot \vec{E} = 4\pi \rho \rightarrow \nabla \cdot (-\nabla \phi - \frac{\partial \vec{A}}{\partial t}) = 4\pi \rho$$

$$-\nabla^2 \phi - \nabla \cdot (\frac{\partial \vec{A}}{\partial t}) = -\nabla^2 \phi - \frac{\partial}{\partial t} \underbrace{\nabla \cdot \vec{A}}_{-\partial_t \phi} = \partial_t^2 \phi - \nabla^2 \phi = 4\pi \rho \rightarrow \square \phi = 4\pi \rho$$

$$J^\mu = (\rho, \vec{J}) \rightarrow \square A^\mu = 4\pi J^\mu$$

أي إشارة مضطرب

$$\eta^{\mu\nu} \partial_\mu A_\nu = 0 \quad \partial_\mu (\eta^{\mu\nu} A_\nu) \quad \text{بجاء } A_\mu, \quad \partial^\mu \text{ بجاء } \partial_\mu$$

معنى جزاء $A_\mu + \partial_\mu \neq$

$$\sum_\nu \eta^{\mu\nu} A_\nu \sim B^\mu \sim A^\mu$$

$$\eta_{\mu\nu} A^\nu = A_\mu$$

$$A^\mu = \eta^{\mu\nu} A_\nu = (\phi, \vec{A})$$

$$\partial_\alpha \partial^\alpha = \eta^{\alpha\beta} \partial_\alpha \partial_\beta = -\partial_0^2 + \partial_i \cdot \partial_i$$

$$\partial_\mu A^\mu = 0$$

$$F_{\mu\nu} \rightarrow F^{\alpha\beta} = \eta^{\alpha\mu} \eta^{\beta\nu} F_{\mu\nu} = \eta F \eta \Rightarrow F^{\alpha\beta} = \partial^\alpha A^\beta - \partial^\beta A^\alpha$$

$$\partial_\alpha F^{\alpha\beta} = \partial_\alpha [\partial^\alpha A^\beta - \partial^\beta A^\alpha] = \partial_\alpha \partial^\alpha A^\beta - \underbrace{\partial_\alpha \partial^\beta A^\alpha}_0 = -\square A^\beta = 4\pi J^\beta$$

$$\partial_\alpha F^{\alpha\beta} = -4\pi J^\beta \rightsquigarrow \frac{\nabla \times \mathbf{B}}{\nabla \cdot \mathbf{E}}$$

$$\partial_\alpha F_{\beta\mu} + \partial_\beta F_{\mu\alpha} + \partial_\mu F_{\alpha\beta} = 0$$

$$\partial_{[\alpha} F_{\beta\mu]} = 0$$

$$\partial_\beta \partial_\alpha F^{\alpha\beta} = -4\pi \partial_\beta J^\beta$$

$$\alpha \leftrightarrow \beta \quad \partial_\alpha \partial_\beta F^{\beta\alpha}$$

$$\rightsquigarrow \partial_\alpha \partial_\beta F^{\beta\alpha} = \partial_\alpha \partial_\beta F^{\alpha\beta} = -\partial_\alpha \partial_\beta F^{\beta\alpha} = 0$$

$$\partial_\alpha \partial_\beta F^{\alpha\beta}$$

$$\Rightarrow \partial_\beta J^\beta = 0 \rightarrow \partial_0 J^0 + \partial_i J^i = \frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0$$

$$F^{\alpha\beta} = \eta^{\alpha\mu} F_{\mu\nu} \eta^{\nu\beta} \rightsquigarrow F^{\alpha\beta} = \begin{bmatrix} 0 & E_1 & E_2 & E_3 \\ -E_1 & 0 & B_3 & -B_2 \\ -E_2 & -B_3 & 0 & B_1 \\ -E_3 & B_2 & -B_1 & 0 \end{bmatrix} = -F_{\alpha\beta}$$

$$E_i = -\partial_0 A_i + \partial_i A_0$$

$$B^i = \frac{1}{2} \varepsilon^{ijk} F_{jk} \quad F_{jk} = \partial_j A_k - \partial_k A_j \quad \text{در ۳ بعد} \quad \begin{matrix} \text{تانسور مرتبه ۲} \\ \text{تانسور مرتبه ۱} \end{matrix}$$

$$\mathbf{f}_i = \rho \mathbf{E}_i + (\mathbf{J} \times \mathbf{B})_i, \quad f_0 dt \propto \mathbf{f}_i \cdot d\mathbf{x}^i = \mathbf{f} \cdot d\mathbf{x} = dW \rightarrow f_0 = \frac{dW}{dt} = \rho = \mathbf{E} \cdot \mathbf{J}$$

$$f_\mu = (-\mathbf{J} \cdot \mathbf{E}, f_i)$$

$$\mathbf{J} \cdot \mathbf{E} = J^i E_i = -J^i F_{i0} + J^0 F_{0i} \Rightarrow f_0 = J^i F_{i0}$$

$$f_\mu \rightsquigarrow f_i = J^0 F_{0i} + \underbrace{B^j \varepsilon_{ijk}}_{F_{jk}} J_k \rightsquigarrow F_{ik} J^k$$

$$f_\mu = J^\nu F_{\nu\mu} \quad f_0 = J^\nu F_{0\nu} = J^0 F_{00} + J^i F_{0i} = -\mathbf{J} \cdot \mathbf{E}$$

$$\mathbf{x}^i \rightsquigarrow \gamma: I \Rightarrow \mathbb{R}^3$$

$$\gamma(t) = (\gamma^1(t), \gamma^2(t), \gamma^3(t))$$

4- بردار سرعت:

$$\underline{\underline{\dot{\gamma}^i}}(\gamma(t)) = \dot{\gamma}^i \rightsquigarrow \dot{\gamma}^i = \dot{x}^i$$

$$\mathbf{x}^i(t) \text{ مولفه } i \text{ تالم ض}$$

$$\dot{\gamma}^i \circ \gamma: t \rightarrow \mathbb{R}^3$$

$$\frac{d\dot{x}^i}{dt} \frac{\partial}{\partial \dot{x}^i} \gamma^j = \frac{d\dot{\gamma}^j}{dt} \rightarrow \frac{d\dot{\gamma}^j}{dt} \cdot \nabla \gamma = \frac{d}{dt} \gamma$$

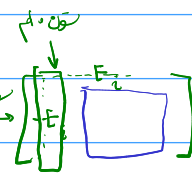
$$x^i = x^i \rightarrow \frac{dx^i}{dt} = f(t) \rightarrow \int dx^i = \int f(t) dt \rightarrow x^i = \int f(t) dt \text{ هم انتگرال}$$

$$\gamma: I \xrightarrow{\varphi} M \quad x^\mu = (t, x^i) \in M \quad u^\mu = \frac{dx^\mu}{d\lambda} \quad x^\mu(\lambda) = x^\mu$$

$$u^\mu = \left(\frac{dt}{d\lambda}, 0, 0, 0 \right) \quad dx^i = 0 \quad \partial\lambda \sim \partial t \rightsquigarrow \partial\lambda = \partial t$$

$$u^\mu = (1, 0, 0, 0) \quad \frac{dx^\mu}{d\lambda} \quad \lambda \text{ اسكن}$$

$$F_{\mu\nu} = \underline{E_\mu u_\nu} - \underline{E_\nu u_\mu}$$

$$F_{i0} = E_i u_0 \quad F_{0i} = -u_0 E_i$$


$$\epsilon_{0123} = +1 \quad \text{مستقر} \quad \epsilon_{\mu\nu\rho\sigma}$$

تجزیه شده

$$*F_{\mu\nu} = -\epsilon_{\mu\nu\sigma\lambda} u^\sigma B^\lambda \rightsquigarrow \text{مگر } \mu, \nu$$

$$\text{ملاحظه } \sigma=0 \quad \lambda=i=1,2,3 \rightarrow \underline{F_{0i}} = -\underline{\epsilon_{0i\nu\lambda} u^\nu B^\lambda}$$

$$F_{\mu\nu} = E_\mu u_\nu - E_\nu u_\mu - \epsilon_{\mu\nu\sigma\lambda} u^\sigma B^\lambda \quad \partial_\mu A_\nu - \partial_\nu A_\mu$$

$$F_{0i} = \cancel{E_0 u_i} - \underline{E_i u_0} - \cancel{\epsilon_{0i\sigma\lambda} u^\sigma B^\lambda}$$

$$S = \int L dt, \quad \delta S = 0 \quad L(x^i, \dot{x}^i)$$

$$L = \int \mathcal{L} dV \rightarrow S = \int \mathcal{L} \frac{dV dt}{d^4 x} \quad d^4 x \rightarrow \text{هولاست}$$

$$S = \int \mathcal{L} \int \underline{d^4 x} = \int \mathcal{L} d^4 x$$

$$L = L(\phi, \partial_\mu \phi)$$

$$\delta S = \delta \int \mathcal{L} d^4 x = \int \underbrace{\delta \mathcal{L}}_{\delta \mathcal{L} = 0} d^4 x = 0$$

$$\frac{\delta \mathcal{L}}{\delta \phi} = \frac{\partial \mathcal{L}}{\partial \phi} + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \underbrace{\delta (\partial_\mu \phi)}_{\partial_\mu (\delta \phi)} = \frac{\partial \mathcal{L}}{\partial \phi} \delta \phi + \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \delta \phi \right) - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right) \delta \phi$$

$$\delta L = \left(\frac{\partial L}{\partial \phi} - \partial_\mu \left(\frac{\partial L}{\partial (\partial_\mu \phi)} \right) \right) \delta \phi$$

$$0 \leftarrow \delta \phi$$

$$\delta S = \int_{\Omega} \left[\frac{\partial L}{\partial \phi} - \partial_\mu \left(\frac{\partial L}{\partial (\partial_\mu \phi)} \right) \right] \delta \phi d^4 x$$

Euler-lagrange

$$\frac{\partial L}{\partial \phi} = \partial_\mu \left(\frac{\partial L}{\partial (\partial_\mu \phi)} \right)$$

$$\frac{\partial L}{\partial x^i} = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}^i} \right)$$

$$\frac{d}{dt} \hookrightarrow \partial_\mu \quad L \hookrightarrow \mathcal{L}$$

$$F^2, F \tilde{F}^2, F^2 \tilde{F}^4, F A^2, A \tilde{F}$$

$$\mathcal{L}(A_\nu, \partial_\mu A_\nu)$$

$$\mathcal{L} = -\frac{1}{16\pi} F_{\mu\nu} F^{\mu\nu} - J^\mu A_\mu$$

$$A_\mu = (-\phi, \vec{A}) \Rightarrow A_\mu = (-\phi, \vec{A}) \quad A^\mu = (\phi, \vec{A}) \quad \eta^{\mu\nu} = \text{diag}(1, -1)$$

$$\frac{\partial \mathcal{L}}{\partial A_\mu} = -J^\mu$$

$$\frac{\partial}{\partial (\partial_\mu A_\nu)}$$

$$\frac{\partial \mathcal{L}}{\partial (\partial_\mu A_\nu)}$$

$$\mathcal{L} = -\frac{1}{16\pi} (\partial_\mu A_\nu - \partial_\nu A_\mu) (\partial^\mu A^\nu - \partial^\nu A^\mu)$$

$$\frac{\partial (\partial_\mu A_\nu)}{\partial (\partial_\mu A_\nu)} = 1$$

$$\frac{\partial (\partial_\nu A_\mu)}{\partial (\partial_\mu A_\nu)} = \delta^\mu_\nu$$

$$F_{\alpha\beta} \frac{\partial (\partial^\alpha A^\beta - \partial^\beta A^\alpha)}{\partial (\partial_\mu A_\nu)} = (\eta^{\alpha\mu} \eta^{\beta\nu} - \eta^{\beta\mu} \eta^{\alpha\nu}) F_{\alpha\beta}$$

$$\frac{\partial F_{\alpha\beta} F^{\alpha\beta}}{\partial (\partial_\mu A_\nu)} = [\eta^{\alpha\mu} \eta^{\beta\nu} - \eta^{\beta\mu} \eta^{\alpha\nu}] F_{\alpha\beta} + \frac{\partial F_{\alpha\beta}}{\partial (\partial_\mu A_\nu)} F^{\alpha\beta} =$$

$$\frac{\partial}{\partial (\partial_\mu A_\nu)} (\partial_\alpha A_\beta - \partial_\beta A_\alpha) = (\delta^\mu_\alpha \delta^\nu_\beta - \delta^\mu_\beta \delta^\nu_\alpha) \Rightarrow$$

$$F^{\mu\nu} - F^{\nu\mu} + F^{\mu\nu} - F^{\nu\mu} = 4 F^{\mu\nu} \rightarrow -\frac{4}{16\pi} \partial_\nu F^{\mu\nu} = -J^\mu$$

$$\partial_\nu \tilde{F}^{\mu\nu} = 4\pi J^\mu$$

$$(\eta^{\nu\alpha} \eta^{\beta\mu} - \eta^{\beta\nu} \eta^{\alpha\mu})$$